

Mitchelle Ayanda Mzolo

Full-Stack Engineer

Gauteng, South Africa | +27 71 849 8044 | ayandamzolo11@gmail.com

Portfolio: ayandamzolo.github.io | LinkedIn: linkedin.com/in/mitchelle-mzolo-4a87b11b5 | GitHub: github.com/ayandamzolo

Professional Summary

Full-stack engineer who builds and secures the systems that move money, documents and messages between companies: bank submission for a mortgage platform serving 6 South African lenders, customer conversations and executive reporting for an internet provider, and pricing logic for a travel booking platform. Strongest at finding failures that do not announce themselves, proving the cause from production evidence, and fixing it once at the point every caller passes through.

Technical Skills

Backend: PHP, Laravel, Python, REST APIs, webhooks, SOAP, queues, multi-tenancy

Data: MySQL, PostgreSQL (materialised views, plpgsql, row level security), Supabase, Redis, MinIO, MeiliSearch

Security: multi-tenant MFA and OTP, WAF and rate limiting, penetration test remediation, audit logging

Testing and delivery: Pest, PHPUnit, Cypress, GitLab CI, GitHub Actions

Frontend: Vue.js, Inertia.js, Ziggy, TypeScript, Tailwind CSS, Blade

Cloud and infrastructure: AWS (ECS, S3, IAM, VPC, Lambda, CloudWatch, Transfer Family), Terraform and CloudFormation (infrastructure as code), Redshift, DMS, Docker, Linux, NGINX, PHP-FPM

AI: Amazon Bedrock

Experience

Junior Software Developer | Blue Bean Software

Nov 2025 - Present

Multi-tenant mortgage origination platform

- Lead developer and platform owner for a multi-tenant platform operating in 2 countries and submitting to 6 South African lenders; primary author of 828 of 968 commits.
- Migrated bank submission to a new vendor with zero resubmissions of in-flight applications by routing each submission on the time it was sent.
- Caught a routing defect in that cutover from production logs (36 of 54 post-cutover submissions misrouted), fixed it with an atomic timestamp and backfill, and consolidated 7 bypassing call sites behind a single resolver.
- Traced 64% of production 5xx errors (1,054 of 1,652 over 14 days) to a nightly cache job running across about 100 domains; removed it and closed an unauthenticated route that let anyone flush caches.
- Diagnosed the out-of-memory crashes behind recurring 502 errors and resized PHP-FPM from 80 workers to 16 with a 256 MB limit, so traffic bursts queue instead of crashing, with NGINX timeouts aligned to the load balancer.
- Built AI document extraction on Amazon Bedrock, now in production, reading payslips and bank statements into application forms; chose the model per document type from measured error rates and validated extraction rules against 2,181 real documents.
- Implemented MFA and OTP verification across every login surface with per-tenant control (55 commits); configured WAF rules and rate limiting and remediated penetration test findings.
- Designed an AWS Transfer Family SFTP architecture with S3, IAM and CloudWatch for partner data exchange, and troubleshot Redshift and DMS synchronisation.
- Wrote 8 production post-mortems and the vendor cutover runbooks; resolved data integrity defects including duplicate reference numbers and financial calculation errors.

Internet provider

- Built a WhatsApp payment recovery bot end to end as sole author (127 commits, 279 automated tests) in Laravel, Inertia and Vue 3, handling out-of-order webhooks, closed conversations and unknown senders.
- Built an executive KPI dashboard (182 commits) in Vue 3, TypeScript and Supabase with CASL permissions, and materialised the reporting views when queries began timing out.

- Extended the commission system with payout logic, a self-service campaign simulator, order import tooling and duplicate order prevention.

Full-Stack Developer | Hit The Road Today

Aug 2024 - Jul 2026

- Sole author of the promotion and discount engine for a travel booking platform: percentage and cash codes governed by 8 eligibility rules, modelled as the code, the rule and the redemption.
- Built refund recalculation for trips modified after a discount was applied, and wrote the business logic specification stakeholders signed off.
- Developed Laravel and Vue.js platform features, handled production debugging and backend optimisation in Docker with Redis and MinIO, and supported deployments and UAT.

Junior Web Developer | Umuzi Academy

Aug 2023 - Aug 2024

- Built frontend applications in HTML, CSS and JavaScript, and REST APIs in Node.js with PostgreSQL, in agile teams.

Projects

- **Persona ZA** (persona-za.netlify.app): synthetic South African identity generator for testing forms, producing valid-format ID numbers with Luhn check digits and 18 fields per record, exported singly or in bulk as JSON or CSV; 8 automated tests validate 2,000 generated identities.
- **BillWell** (billwell.netlify.app): invoice generator that renders styled A4 PDFs, running a slimmed Chromium build on serverless hosting with a smoke test confirming valid PDF output.
- **ModernTsonga** (moderntsonga.netlify.app): reference site for Xitsonga phrases, proverbs and oral tradition.

Leadership

- Mentor, Gaming 4 Girls Hackathon (Umuzi and UNICEF, 2023): guided a team to 2nd place.

Education and Certifications

- AWS Certified Cloud Practitioner (CLF-C02), 2026
- University of South Africa, National Diploma: Information Technology, 2025 - Present
- Umuzi Academy, National Certificate: IT Systems Development (NQF 5), 2023 - 2024
- Tshwane University of Technology, National Diploma: Information Technology (incomplete)